Krishnaa **Vadivel** | PhD Research Engineer | Computational Physics | EE Systems | Scientific ML

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Profile

Broad-profile electrical engineering PhD researcher with training and project work spanning computational physics, semiconductor and photonics theory, scientific machine learning, embedded systems, digital design, and high-performance scientific software. The core strength is connecting mathematical models, physical intuition, and implementation detail across research and engineering settings.

Research Experience

Yale University, Electrical Engineering
Research Assistant / PhD Researcher *2021–present*
- Develop first-principles and many-body approaches for self-trapped excitons, excitonic polarons, and electron-phonon coupled systems, connecting DFT, DFPT, excited-state theory, and effective Hamiltonian viewpoints.
- Study coherent-state and path-integral formulations for interacting quantum systems, including time-dependent and self-localized phenomena relevant to condensed-matter and materials problems.
- Build distributed scientific-computing workflows in Python and C++ using MPI, OpenMP, PETSc, and SLEPc for large simulation and linear-algebra workloads.
- Port and optimize workloads for accelerator-enabled HPC systems with CUDA and ROCm, with practical experience on Frontier, Perlmutter, Frontera, and related clusters.
- Explore ML accelerators for scientific computing, including equivariant graph neural networks for molecular-property prediction and agentic retrieval workflows for generating structured computational-science inputs.
- Support embedded-systems instruction and hardware-oriented lab work through teaching-assistant responsibilities tied to firmware, debugging, and instrumentation.
- Co-authored a 2025 *Journal of the American Chemical Society* paper on ultrafast self-trapped exciton formation in lead-free double halide perovskites and presented related APS talks in 2024, 2025, and 2026.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer *2019–2021*
- Built CUDA-based data-processing pipelines and custom GPU-enabled engineering tooling for high-volume acoustic waveform datasets used in deployed well-logging tools.
- Developed embedded software and digital logic for sensing, logging, and control functions using Lattice FPGA, PIC32, dsPIC, STM32, and C8051-based systems.
- Supported board- and system-level engineering tasks including oscilloscope-based debugging, integration testing, field validation, and selective PCB modifications in Altium.
- Contributed host-side tooling with C#, ASP-style services, and Microsoft SQL-backed database workflows supporting engineering operations and data handling.

FindLight
Photonics Technical Writer *2018–2019*
- Produced technical articles and engineering-facing content covering lasers, optics, photonics components, and optoelectronic systems.

Selected Projects

Distributed Exciton-Phonon and Materials Simulation

- Developed scalable first-principles and many-body workflows for excited-state materials problems, emphasizing reusable HPC infrastructure and theory-to-code translation.

Semiconductor Device Simulation
- Built numerical treatments of electrostatics and transport equations, including Poisson, drift-diffusion, and pn-junction-style models linking semiconductor theory with simulation practice.

Equivariant Molecular Machine Learning
- Implemented graph-based and symmetry-aware ML models for molecular-property prediction and investigated their role in accelerating scientific workflows.

FPGA AI Accelerator and Digital Design Work
- Implemented Verilog-based accelerator and processor projects to demonstrate datapath design, verification workflows, and hardware-oriented compute thinking.

Embedded Robotics and Human-Interface Projects
- Built robotics and embedded-system class projects including a surface-EMG robotic hand that was later published in Circuit Cellar, plus maze-navigation work with wireless coordination.

Integrated Photonics and FDTD Modeling
- Simulated band-pass and resonator-style photonic structures with Meep-based FDTD workflows for integrated-optics exploration.

Agentic Scientific Automation
- Built retrieval-driven tooling that reads software documentation and research papers to generate structured inputs and workflow scaffolds for computational science.

Education

Yale University
PhD, Electrical Engineering *2021–present*
Ongoing work spanning computational materials, many-body theory, semiconductor-relevant modeling, and scientific software.

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*
Included stochastic-computing-related work, device-oriented EE topics, and advanced design coursework.

Cornell University
BS, Electrical and Computer Engineering *2013–2017*
Included embedded systems, robotics, controls, and strong math/physics/computing preparation.

Selected Publications

2025: de Paula, A. M., Li, S., Hou, B., Vadivel, S., Teles-Ferreira, D. C., Iudica, A., Kabacinski, P., Hosseini, H., McArthur, J., Cerullo, G., et al. (2025). Time-domain observation of ultrafast self-trapped exciton formation in lead-free double halide perovskites. *Journal of the American Chemical Society, 147*(32), 28923–28931.

Selected Conference Presentations

2026: Vadivel, S., Grande, R. D., Strubbe, D. A., & Qiu, D. Y. (2026). First-principles study of exciton-polarons and self-trapped excitons. *APS Global Physics Summit 2026*.

2025: Vadivel, S., Qiu, D. Y., Grande, R. D., & Strubbe, D. A. (2025). First-principles study of self-trapped exciton formation in double perovskites. *APS Global Physics Summit 2025*.

2024: Vadivel, S., & Qiu, D. (2024). First-principles study of self-trapped exciton formation in double perovskites using excited state forces. *APS March Meeting Abstracts 2024*, F01–007.

Selected Coursework

Mathematics: Differential Geometry; Abstract Algebra; Honors Mathematical Analysis II

Computer Science: Theoretical Computer Science; Probabilistic Machine Learning

Physics: Graduate Quantum Mechanics I–II; Solid State Physics I–II; Many-Body Quantum Physics

Electrical Engineering: Semiconductor Physics; Thin Film Science; Lasers and Optoelectronics; Integrated Photonics; VLSI Design; Analog VLSI Design; Mathematics of Signals and Systems; Advanced Embedded Systems

Technical Skills

Languages: Python, C, C++, CUDA, JavaScript, Bash, PowerShell, C#

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, mpi4py, petsc4py, slepc4py, NumPy, pandas, SciPy, SymPy, matplotlib

Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks, scientific ML workflows

Hardware: Verilog, Lattice FPGA, PIC32, dsPIC, STM32, C8051, sensor interfaces, PCB bring-up, oscilloscope debugging

Software / Systems: ASP.NET, Microsoft SQL Server, Linux command line, Docker, Docker Compose, Kubernetes, gcloud

HPC Platforms: Frontier, Frontera, Perlmutter, and related national-lab or university HPC environments